

Laporan Praktikum WEEK 11 Jaringan Komputer

Modul 11 DHCP

Tujuan :

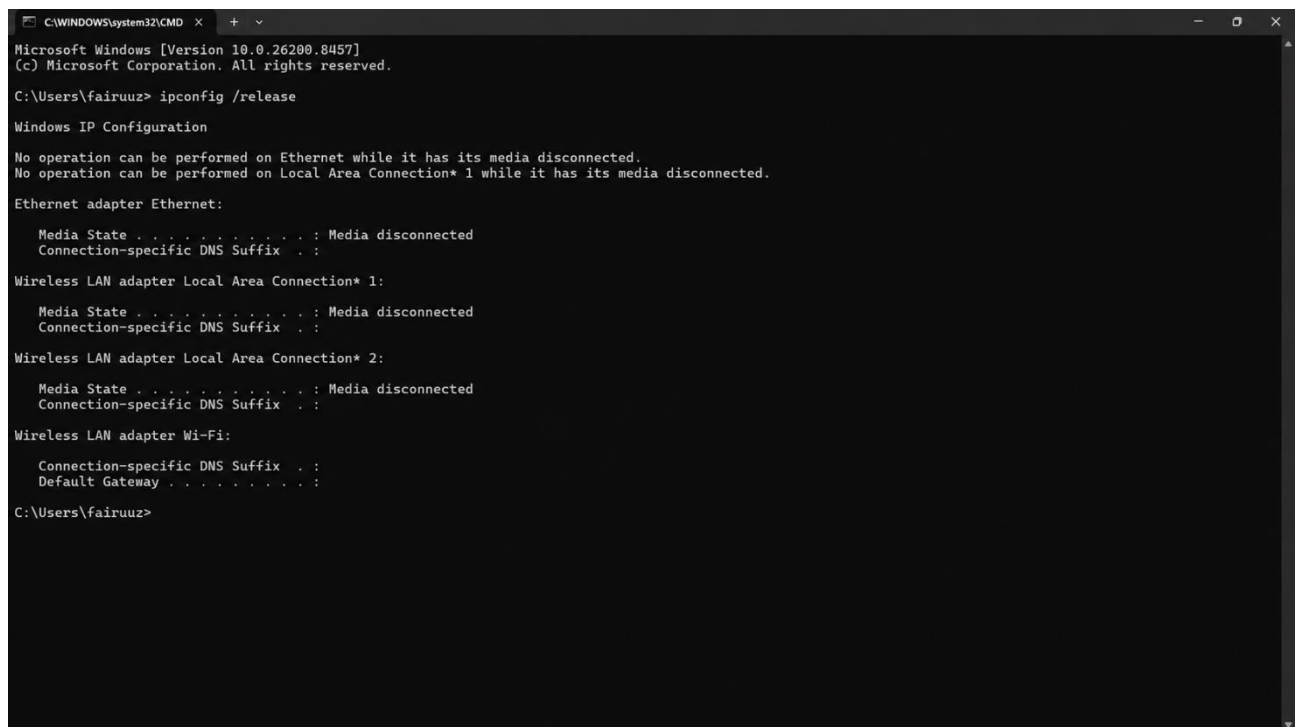
1. Mahasiswa dapat menginvestigasi cara kerja protokol DHCP menggunakan Wireshark

Pengantar

Di modul ini, kita akan melihat sekilas Dynamic Host Configuration Protocol, DHCP. Ingat bahwa DHCP digunakan secara luas di perusahaan, universitas dan LAN kabel dan nirkabel jaringan rumah untuk secara dinamis menetapkan alamat IP ke host, serta untuk mengkonfigurasi informasi konfigurasi jaringan lainnya. Seperti yang telah kami lakukan di lab Wireshark sebelumnya, Anda akan melakukan beberapa tindakan di komputer Anda yang akan menyebabkan DHCP beraksi, dan kemudian menggunakan Wireshark untuk mengumpulkan dan kemudian jejak paket yang berisi pesan protokol DHCP.

Mengumpulkan Jejak Paket :

1. IPCONFIG/RELEASE :



```
C:\WINDOWS\system32\CMD X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz> ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Default Gateway . . . . . :

C:\Users\fairuuz>
```

2. IPCONFIG/RENEW :

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26280.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Wi-Fi while it has its media disconnected.

C:\Windows\System32>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::3932:75d0:5e6f:8c31%12
    IPv4 Address. . . . . : 192.168.1.53
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Windows\System32>
```

3. DHCP REQUEST :

The image shows a Wireshark packet capture of a DHCP request. The packet list pane displays several packets, with packet 1223 selected. The packet details pane shows the structure of the DHCP request, and the packet bytes pane shows the raw data.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
2	0.029406400	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xb195949f
79	2.595543800	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xb195949f
95	2.747688400	192.168.1.1	255.255.255.255	DHCP	357	DHCP ACK - Transaction ID 0xb195949f
1223	19.5666038	192.168.1.53	192.168.1.1	DHCP	358	DHCP Request - Transaction ID 0xb26ae3d3
1224	19.6108185	192.168.1.1	192.168.1.53	DHCP	357	DHCP ACK - Transaction ID 0xb26ae3d3

Packet 1223 Details:

- Frame 1223: Packet, 358 bytes on wire (2864 bits), 358 bytes captured (2864 bits) on interface 0
- Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: zte_23:db:9a (bc:bd:84:23:db:9a)
- Internet Protocol Version 4, Src: 192.168.1.53, Dst: 192.168.1.1
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Source Port: 68
- Destination Port: 67
- Length: 324
- Checksum: 0x4f0f [unverified]
- [Checksum Status: Unverified]
- [Stream index: 61]
- [Stream Packet Number: 1]
- [Timestamps]
- UDP payload (316 bytes)
- Dynamic Host Configuration Protocol (Request)
- Message type: Boot Request (1)
- Hardware type: Ethernet (0x01)
- Hardware address length: 6
- Hops: 0
- Transaction ID: 0xb26ae3d3
- Seconds since boot: 0

Packet Bytes:

0000 bc bd 84 23 db 9a 24 b2 b9 af cf 7d 08 00 45 00 ... # \$... } E

0010 01 58 f2 45 00 00 00 11 c3 c8 c0 a8 01 35 c0 a8 ... X E ... 5

0020 01 01 00 44 00 43 01 44 4f 0f 01 01 06 00 b2 6a ... - D - C - D 0 ... j

0030 e3 d3 00 00 00 00 c0 a8 01 35 00 00 00 00 00 00 ... 5

0040 00 00 00 00 00 24 b2 b9 af cf 7d 00 00 00 00 00 ... \$... }

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...

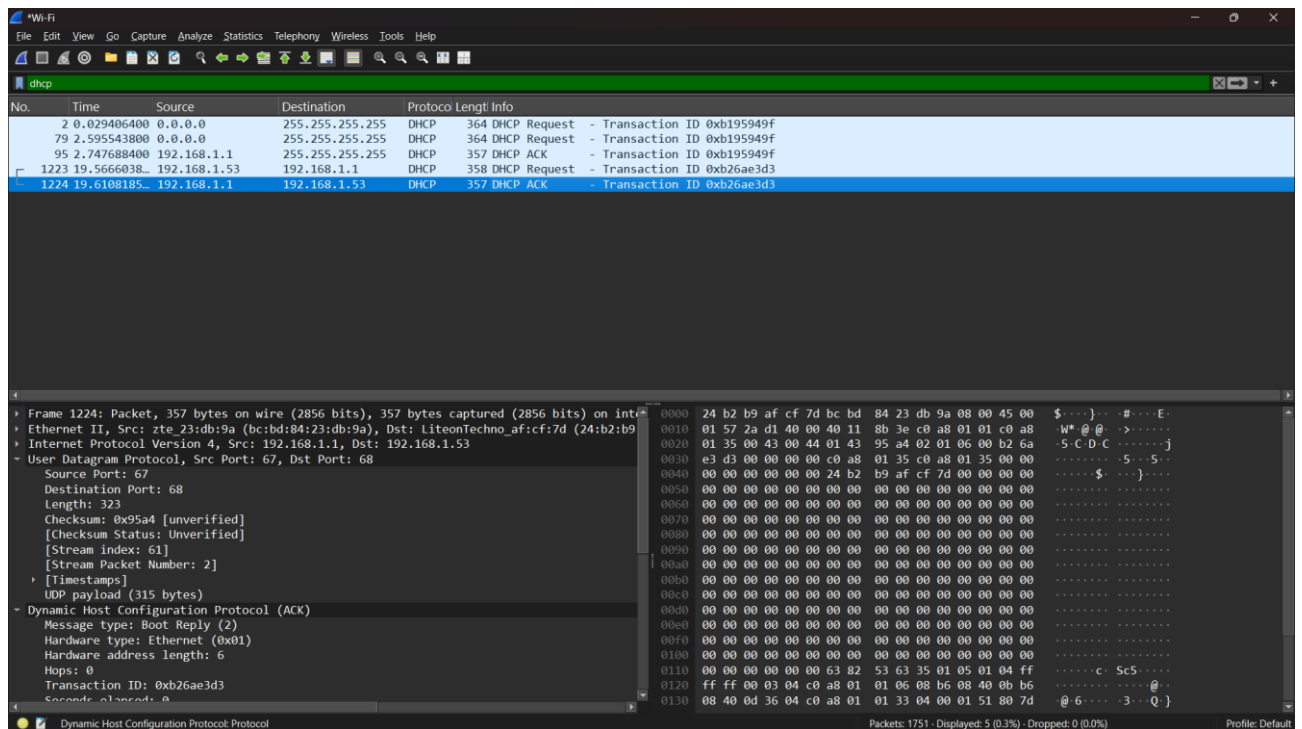
0110 00 00 00 00 00 63 82 53 63 35 01 03 3d 07 01 ... c \$ c5 ...

0120 24 b2 b9 af cf 7d 0c 0f 4c 41 50 54 4f 50 2d 46 ... \$... } LAPTOP-F

0130 4a 37 51 37 33 36 31 51 12 00 00 00 4c 41 50 54 ... 7Q7361Q ... LAPT

Packets: 1751 - Displayed: 5 (0.3%) - Dropped: 0 (0.0%) Profile: Default

4. DHCP ACK :



Pada percobaan DHCP di Windows, paket yang tertangkap hanya DHCP Request dan DHCP ACK. Hal ini terjadi karena sistem masih memiliki lease IP sebelumnya sehingga proses DHCP berlangsung dalam mode renewal, bukan initial discovery penuh.